

ATC Automation Test Framework(AATF) Sharing



BBF-412-1 Li Liying

BMW Brilliance Automotive

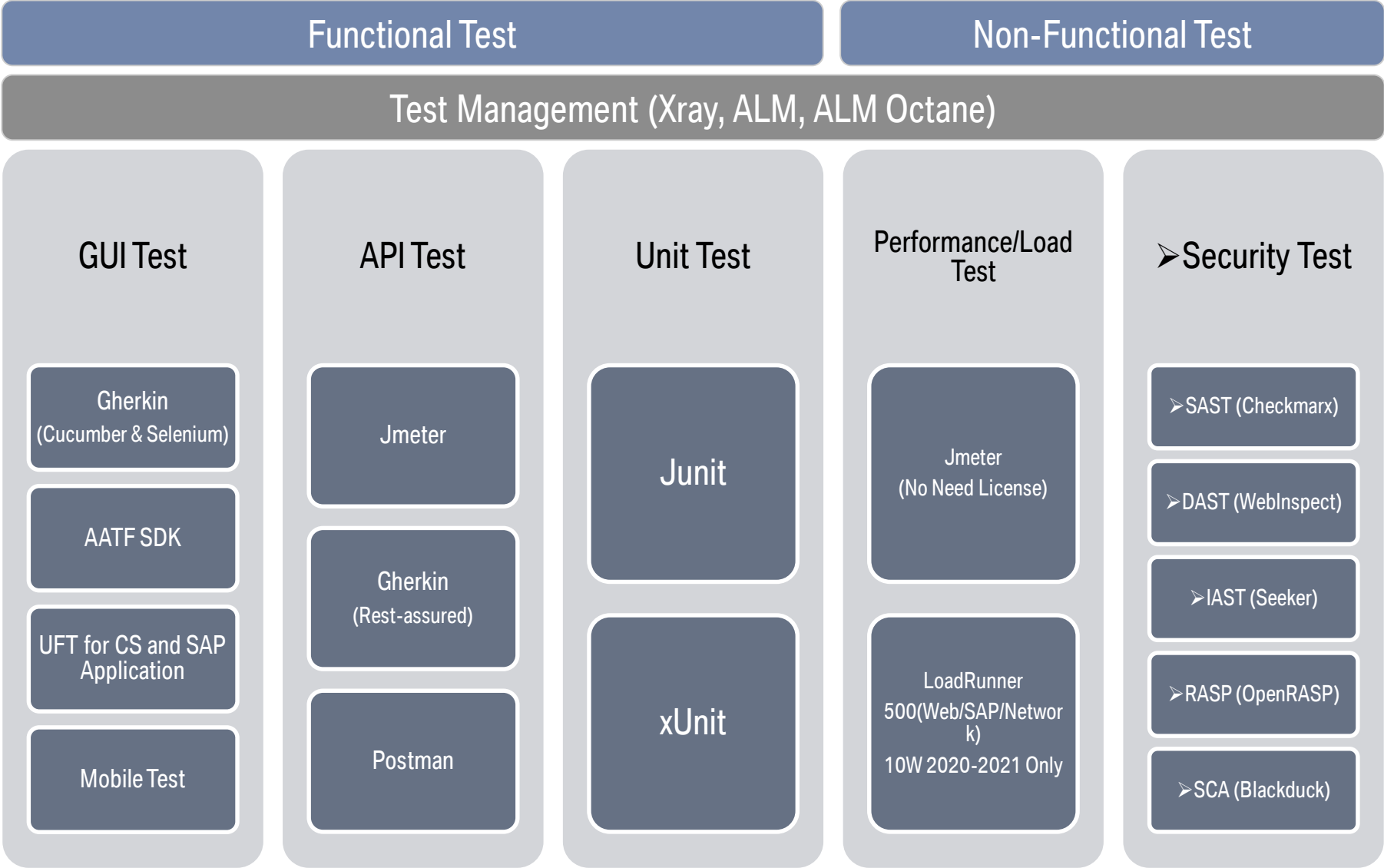
悦 创造奇迹



之诺
ZINORO

WHAT CAN WE DO?

ATC AUTOMATION TEST FRAMEWORK



AUTOMATION TEST TOOLS IN BBA ATC

1. Test Management Tools



2. Unit Test

3. Functional Test Tools in ATC(ATC Automation Test Framework)

➤ B/S Application Automation test(Open Source)



➤ C/S OR C/S + B/S+ Mobile (ATC Provide License)



4. Non-Functional Test

➤ Performance/Load Test



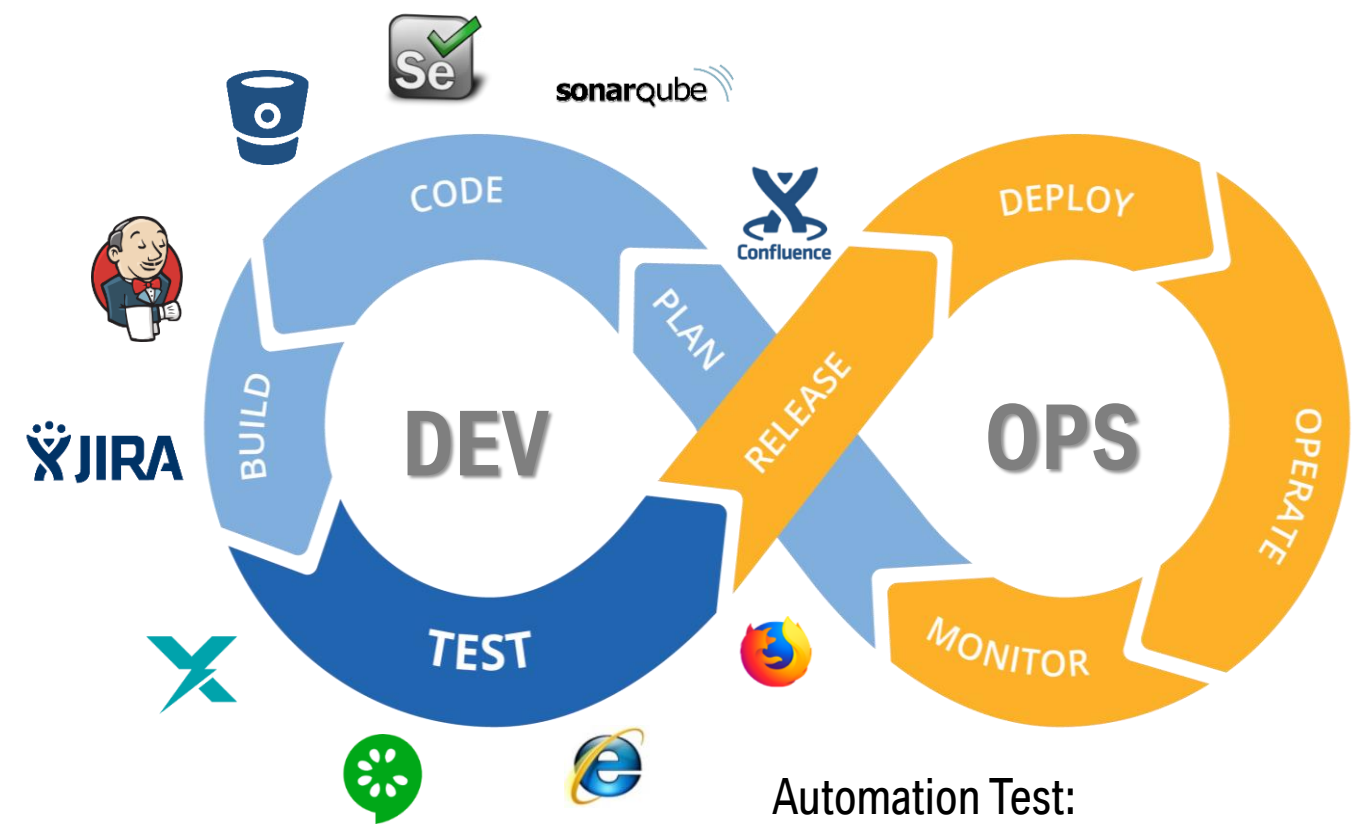
➤ Security Test



AGENDA

- Automation Basic Information
- What Tools Are We Using?
- Automation Test Execution Flow
- Write The Tests

TEST AUTOMATION



Manual Test:

- ✗ More and More Test Case will be run before each release.
- ✗ More and more release is coming for each product, this will be require more and more regression test.
- ✗ The application changed more and more complex, it's hard to ensure the quality.

Automation Test:

- ✓ Write the automation test case by using Gherkins, kinds of natural language.
- ✓ Automation test case will be run before each release automatically.
- ✓ Using Xray as the test management tool to trigger the test, record the test execution, generate the test report, all of this are generate automatically.

STATUS @ AUTOMATION

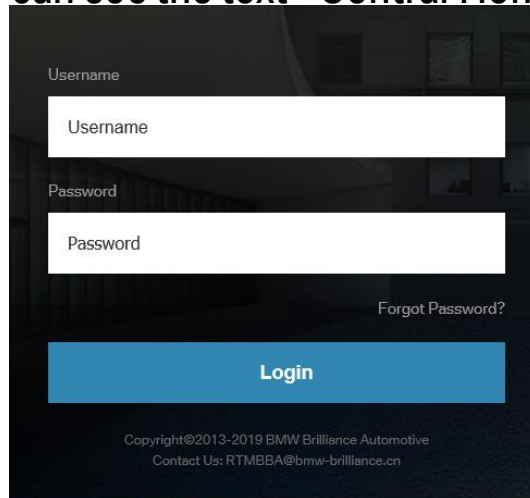
DEMO: AUTOMATION TEST TO CONFIRM THE LOGIN IS SUCCESSFUL

User Story:

As a user,

I want to login RTM with the correct username and password.

So that I can see the text “Central Homologation & Regulation Platform” is displayed



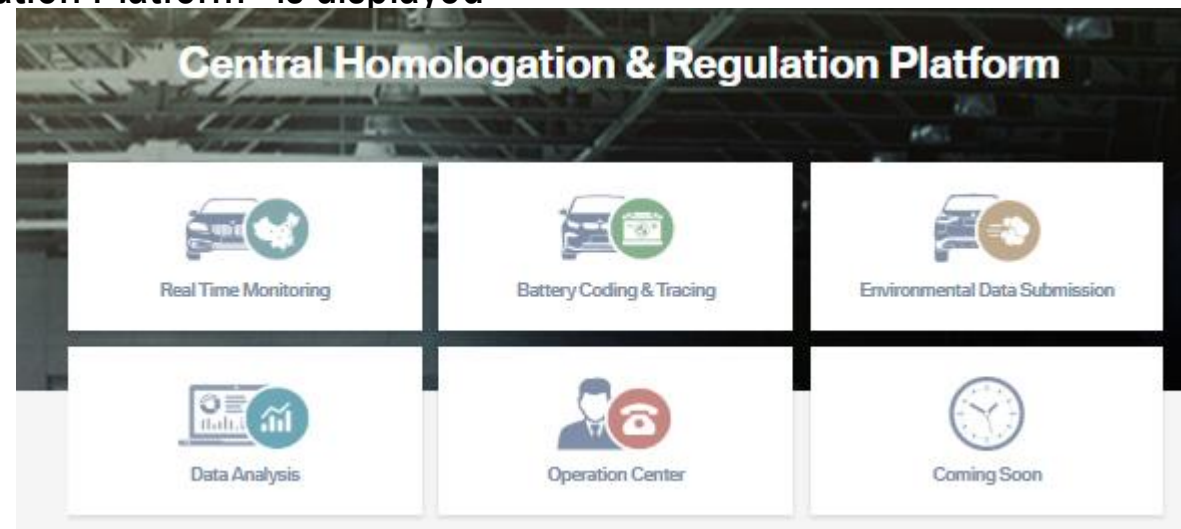
Username

Password

[Forgot Password?](#)

Login

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Contact Us: RTMBBA@bmw-brilliance.cn



Given I navigate to "https://homologation-int.bmw-brilliance.cn/central_platform/login"

And I maximize browser window

When I enter "liliying" into input field having id "username"

And I enter encrypt password "DuUhty/i263ugtdske0d6gysKFAlgcjA1apOVUoelDo=" into input field having id "password"

And I click on element having id "loginBtn"

And I click on link having text "EN"

Then element having class "platform-name" should have text as "Central Homologation & Regulation Platform"

And I take screenshot

And I click on link having text "Exit"

C/S APPLICATION

DEMO SAP CREATE STOCK ORDER

- Story: As a user, I want to navigate to the 'Create the SO' page in SAP CQI system, then make sure I can input the correct value as below.
- T-Code: VA01
 - 订单类型(AUART) : SPSO
 - 分销渠道(VTWEK) : G1
 - 产品组(SPART) : G1
- 售达方(KUNNR) : 28426
- 采购订单编号(BSTKD) : Test RPA XXX
- 物料(MATNR) : 85452547376
- 数量(KWMENG) : 10

创建销售订单: 初始屏幕

销售 项目总览 订货方 特性

订单类型 **SPSO** P SY Stock Order

组织数据

销售组织	SY02	Spare Parts
分销渠道	G1	WIT Dealer
产品组	G1	BMW-Cars
销售部门		
销售组		

创建 P SY Stock Order: 概览

订单

P SY Stock Order 净价值 9,000.00 RMB

售达方 28426 广东粤宝汽车销售服务有限公司 / 广州市白云区白云大道南1115号 / 5104...

送达方 28426 广东粤宝汽车销售服务有限公司 / 广州市白云区白云大道南1115号 / 5104...

采购订单编号 Test RPA 采购订单日期 2019-11-11

销售 项目总览 项目细节 订货方 采购 装运 快速数据输入 拒绝原因

请求交货日期 D 2019-11-11 交货工厂

☐ 全部交货 总重量 122.850 KG

交货冻结 no block 业务量 9.710 HL

出具发票冻结 定价日期 2019-11-11

付款条件 Y112 Payment in 30 Days

国际贸易条件 CPT Carriage Paid To

订单原因

销售范围 SY02 / G1 / G1 Spare Parts, WIT Dealer, BMW-Cars

全部项目

项目	物料	订单数量	计...单位	类别	工厂	描述	拒绝原因
10	85452547376	10	00	SPSO	SY03	Pirelli Cinturato P7 r-f	

UFT MOBILE DEMO

The screenshot displays the UFT Mobile Demo application interface. The top menu bar includes File, Edit, View, Search, Design, Record, Run, Resources, ALM, Tools, Window, and Help. Below the menu is a toolbar with various icons for file operations and testing. The main window is titled "UFT Report : GUITest11 - Res12" and contains a script editor with the following code:

```
1 'Device("Device").App("Uat").MobileEdit("MobileEdit").Tap
2 'Device("Device").EnterKeys 0,"TA.TA"
3 'Device("Device").App("Uat").MobileEdit("MobileEdit_2").Tap
4 'Device("Device").EnterKeys 0,"JoyLearning@2019"
5
```

Below the script editor is a "Data" table with the following content:

A1
A
P
2
3
4

The bottom status bar shows "Ready" and "In 5 col 1 ch 1". The Windows taskbar at the bottom indicates the time is 10:34 AM on 3/18/2021.

Auto Test As Monitor



- ✓ Monitor application's core functions.
- ✓ Monitor APIs Status.
- ✓ Auto-notification.
- ✓ Auto-save error log as evidence.

Sample Application:

RTM, Jira, Confluence, HR SSC Reporting, IPAM, ProSec-Net NEXT Self-Service Portal.

Auto Test As 'Robot'

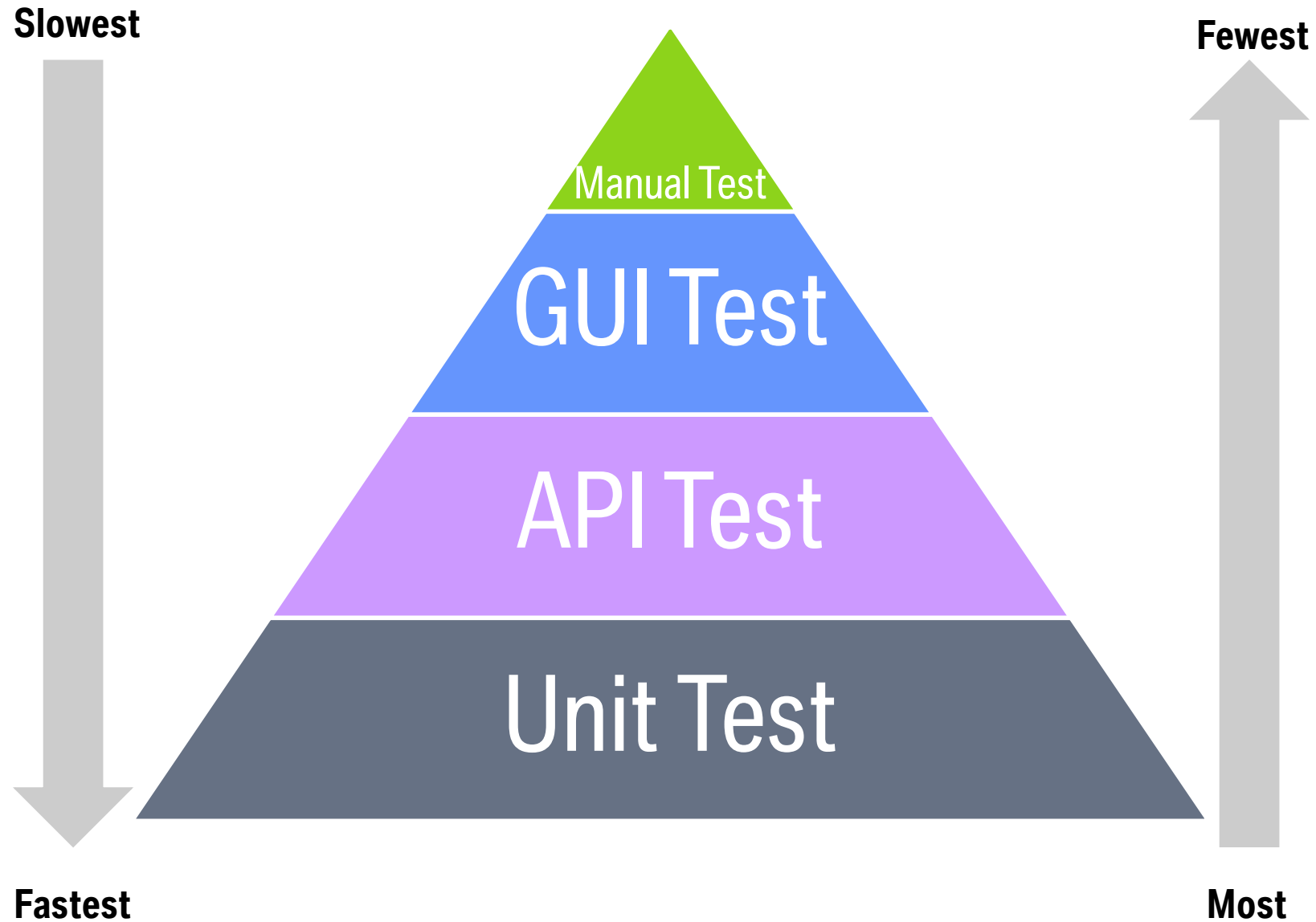


- ✓ Simulate real user actions.
- ✓ Simulate operation actions.
- ✓ Support all types of application(B/S,C/S,APIs).

Sample Application:

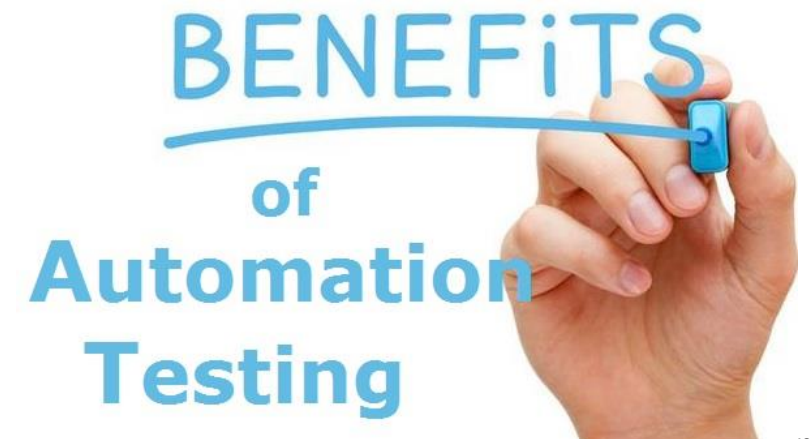
Global X Auto Create User, Space, Technical User and Group Space.

TEST PYRAMID IN AGILE



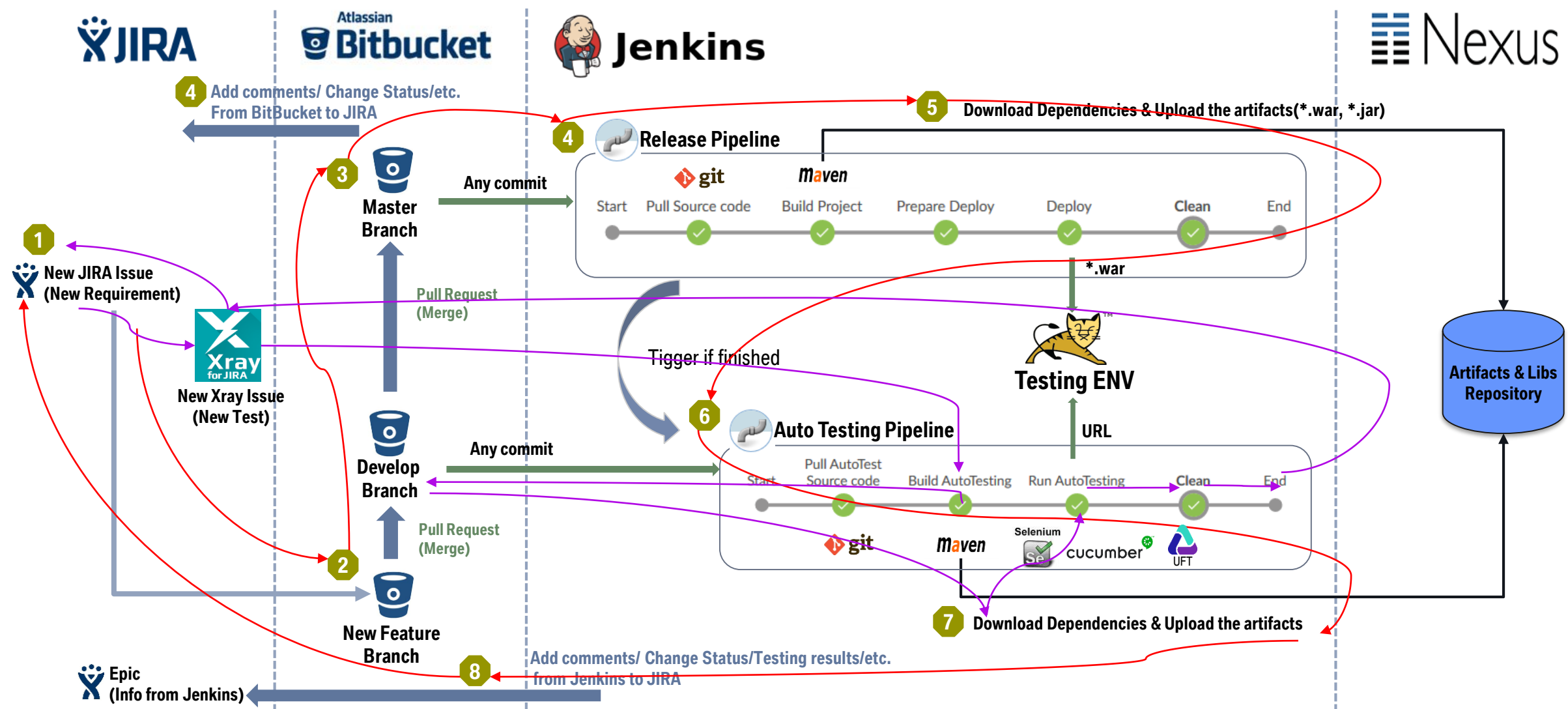
WHY WE NEED AUTOMATION TEST

1. Continuous release requires more and more regression test before each release.
2. Automation helps increase Test Coverage.
3. As agile development process, the regression test would be done very frequently (e.g. by week).
4. The UI logic is getting more complex. And may test cross-platform(Linux, Mac, windows), cross-browser(Firefox, Chrome, IE).
5. One testing team may responsible for multiple projects, so the limited resource is one of the important issue.
6. Manual Testing of all work flows, all fields , all negative scenarios are time and cost consuming.
7. Automation does not require Human intervention. You can run automated test unattended (overnight).
8. Automation increases speed of test execution.



STATUS @ CONTINUES INTEGRATION

DEMO: AUTOMATION PLANNING TO DEPLOYMENT



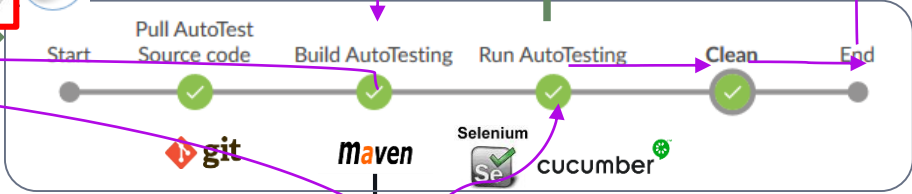
DEMO: AUTOMATION TEST PLAN RUN ON SCHEDULED



Tigger if finished



Auto Testing Pipeline



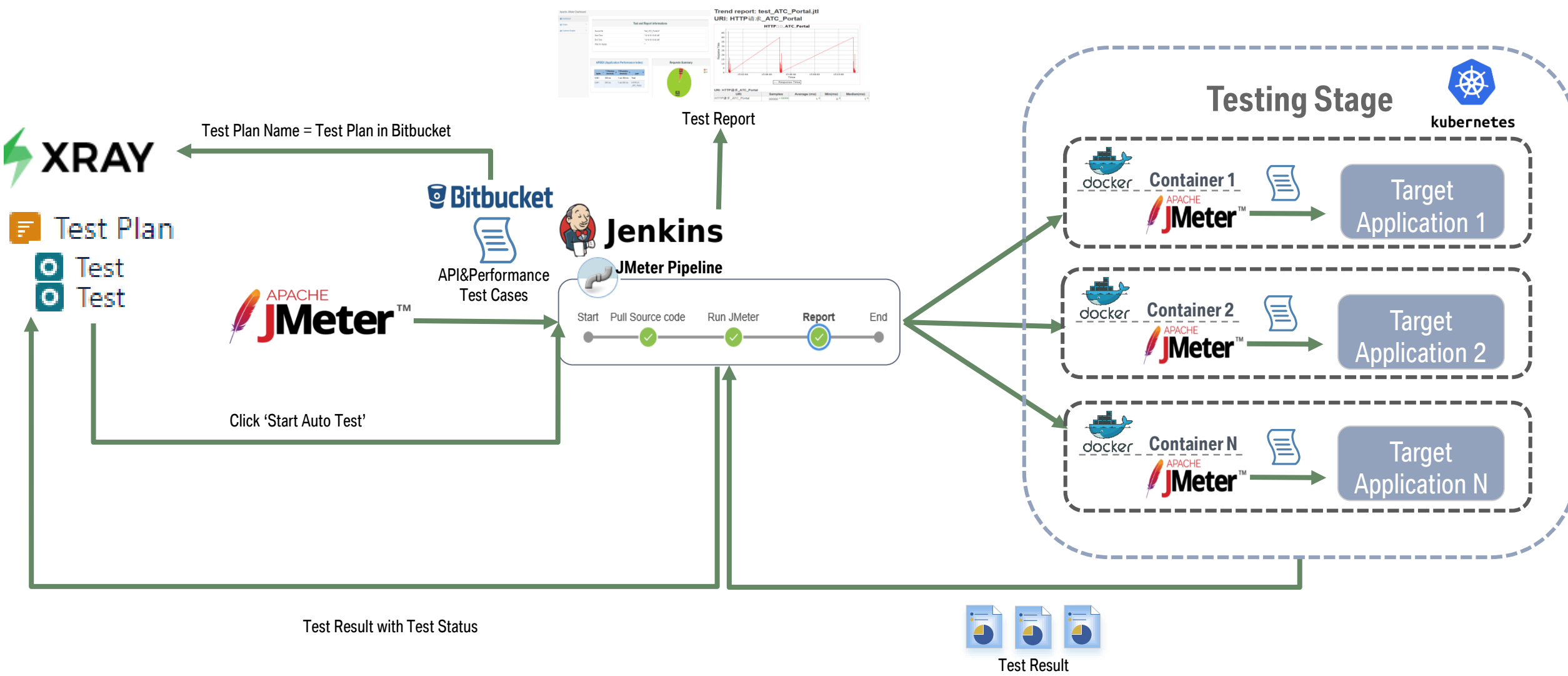
URL

Download Dependencies & Upload the artifacts



DEMO @PERFORMANCE & API TESTING QA

JMETER SERVICE FOR DEVOPS API & PERFORMANCE TESTING



WHAT TOOLS?



Jenkins



Git Client



maven



REST-assured

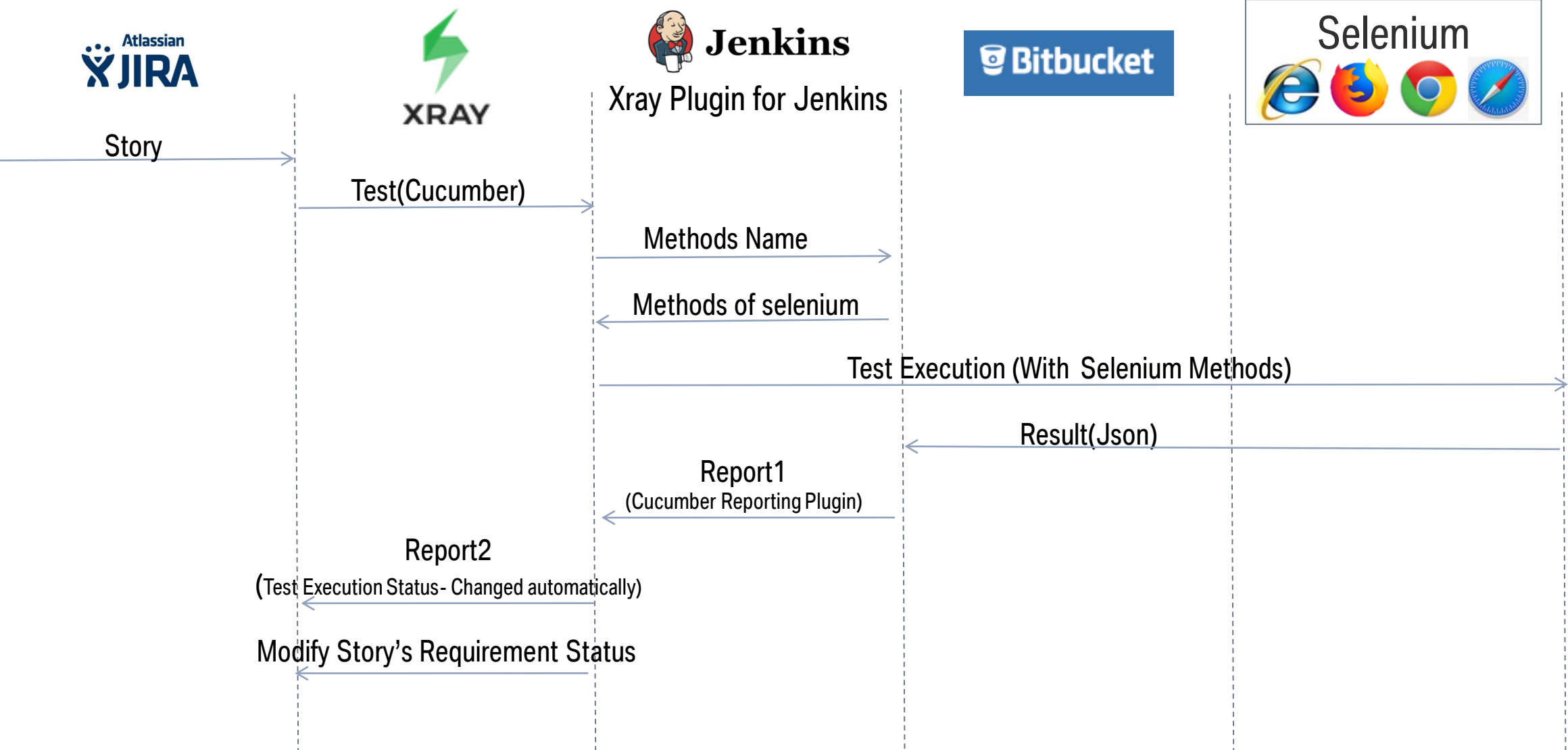
cucumber



Nexus



TEST CASE EXECUTE FLOW- DATA FLOW

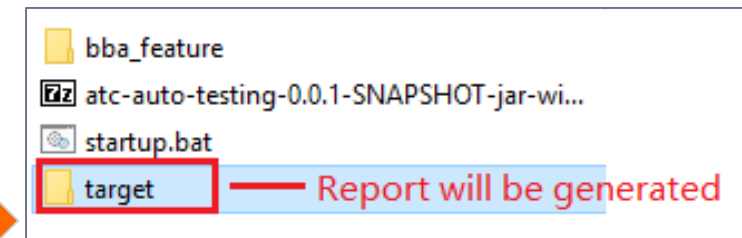
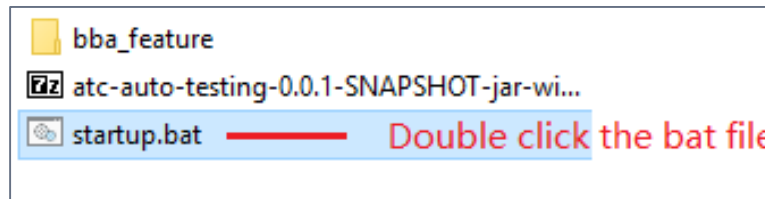
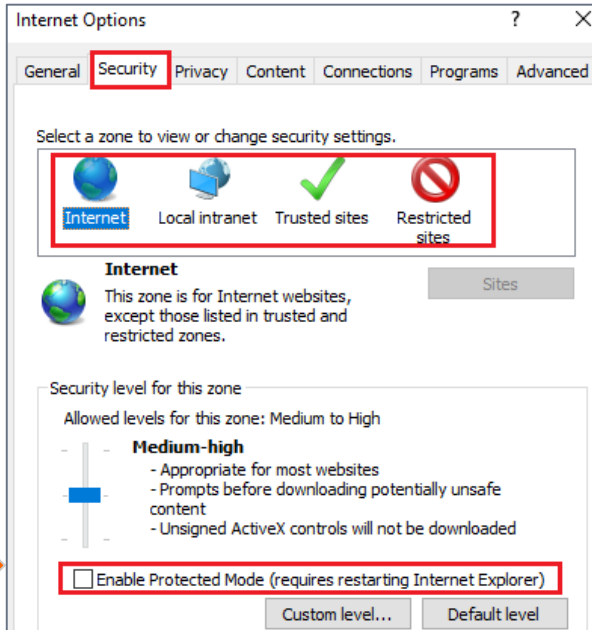
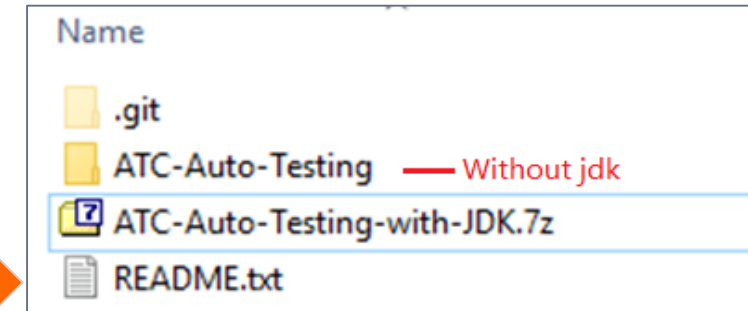
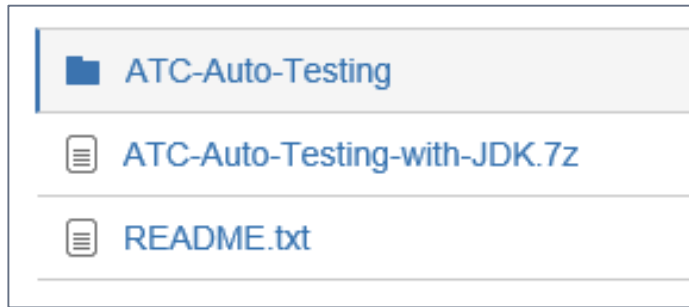


WRITE THE TESTS

1. Environment set-up.
2. Learn the Keywords
3. How to write the test details?
4. Xpath introduction.

TEST ENVIRONMENT SET-UP

Open Link: <https://atc.bmw-brilliance.cn/bitbucket/users/u0028032/repos/atc-autotesting-framework/browse>



Please open: <target\cucumber-html-reports\overview-features.html>

Automation 'Local Run' quick set-up

<https://confluenceit.bmw-brilliance.cn/pages/viewpage.action?pageId=29530086>

TEST

Keywords

Feature: Check the ATC home page

As a user, I want to open BBA home page, and navigate to ATC homepage.

Background:

Given I navigate to "http://www.bba/bba_me/Pages/Home.aspx#"

And I maximize browser window

Scenario: As a user, I want to see the 'ATC' as the page title for the ATC homepage.

When I click on link having text "Overview"

And I click on link having text "IT"

And I wait for 2 sec

And I click on link having text "ATC"

And I switch to new window

And I wait for 2 sec

Then I should see page title having partial text as "ATC"

Summary for Feature/Scenario

Cucumber Test Details



BBAHomePage.feature

HOW TO WRITE THE TEST CASE?

Here is the list of keywords that Gherkin(Cucumber) supports:

- **Feature:** Each Gherkin file begins with a Feature keyword. Feature defines the logical test functionality you will test in this feature file. The idea of having a feature file is to put down a **summary** of what you will be testing.
- **Background:** Background keyword is used to define steps which are **common** to all the tests in the feature file.
- **Scenario:** Each Feature will contain some number of tests to test the feature. Each test is called a Scenario and is described using the 'Scenario'.
- **Given:** This keyword defines a **precondition** to the test.
- **When:** This keyword defines the **test action** that will be executed. By test action we mean the user input action.
- **Then:** This keyword defines the Outcome of previous steps. The **expected result** for your current scenario.
- **And:** This keyword is used to **add conditions to your previous steps**.
- **But:** This keyword is used to add **negative** type comments.
- ***:** This keyword is very **special**, it defies the whole purpose of having Given, When, Then and all the other keywords. Basically Cucumber doesn't care about what Keyword you use to define test steps, all it cares about what code it needs to execute for each step.

Guideline and Samples:

<https://confluenceit.bmw-brilliance.cn/display/ATCS/How+to+create+the+Gherkin%28Cucumber%29+Test>

HOW TO WRITE THE TEST DETAILS.

Example1

//Step to navigate to specified URL

@Then("^I navigate to \"([^\"]*)\"\$")

Example:

Given I navigate to "**http://160.46.78.46:9620/rtm/a/login**"

Example2

@Then("^I enter \"([^\"]*)\" into input field having (.+) \"([^\"]*)\"\$")

"([^\"]*)" is the value you want to entered

(.+) is the locator type: **index, id, name, class, xpath, css**

"([^\"]*)" is the locator value: **index value; id value; name value; class value; xpath value; css value**

Example: Then I enter "**liliying**" into input field having **id "username"**

Example3

// click on web element

@Then("^I click on element having (.+) \"(.*)\"\$")

(.+) is the locator type: **index, id, name, class, xpath, css**

"(.*)" is the locator value: **index value; id value; name value; class value; xpath value; css value**

Example: When I click on element having **id "loginBtn "**

XPATH

➤What is XPath?

XPath (XML Path Language) is a query language for selecting nodes from an XML document.

➤Xpath Terminology 术语

节点(Node)

```
<?xml version="1.0" encoding="ISO-8859-1"?>
```

```
<bookstore> 开始标签
```

```
<book>
```

```
<title lang="en">Harry Potter</title>
```

```
<author>J K. Rowling</author>
```

```
<year>2005</year>
```

```
<price>29.99</price>
```

```
</book>
```

```
</bookstore>结束标签
```

Relationship of each node(节点之间的关系)

- ✓ **父(Parent):** book 是 title, author, year 以及 price 的父;
- ✓ **子(Children):** title, author, year 以及 price 都是 book 的子;
- ✓ **同胞(Sibling):** title, author, year 以及 price 都是同胞;
- ✓ **先辈(Anccestor):** title 的先辈是 book 和 bookstore;
- ✓ **后代(Descendant):** bookstore 的后代是 book, title, author, year 以及 price .

XPATH

➤ Xpath axis specifiers

/	The / character indicates the root node(从根节点选取).
//	Locate the current node in xml, you don't need to think about the detail location for that node. (从匹配选择的当前节点选择文档中的节点，而不考虑它们的位置)
.	The . character indicates the current node(选取当前节点).
..	The . character indicates the current node's Parent node(选取当前节点的父节点).
@	Indicated the attribute选取属性。

Example1 :

<A>

<C>fff</C>

<D>ggg</D>

<E>hhh</E>

1. /A/B/C 一种是绝对路径（以第一个标签为参照物）

2. //C 相对路径（以其他已知的标签为参照物）

Example2 : <http://160.46.78.46:9620/rtm/a/login?>

Xpath: html/body/div/div/img

```
<html xmlns="http://www.w3.org/1999/xhtml">
  <head>
  <body>
    <div id="login_main" class="login_main" style="margin-top: 20px;">
      <div id="login_logo" class="login_logo">
        
      </div>
```

Xpath: //img

```
<html xmlns="http://www.w3.org/1999/xhtml">
  <head>
  <body>
    <div id="login_main" class="login_main" style="margin-top: 20px;">
      <div id="login_logo" class="login_logo">
        
      </div>
```

XPATH COMMON FUNCTION

1.contains(): //div[contains(@id,'logo')],表示选择id中包含有'logo'的div节点

```
Top Window ▾ Highlight XPath: ▾ //div[contains(@id,'logo')]
<document>
  <html xmlns="http://www.w3.org/1999/xhtml">
    <head>
    <body>
      <div id="login_main" class="login_main" style="margin-top: 20px;">
        <div id="login_logo" class="login_logo">
```

2.text(): 由于一个节点的文本值不属于属性.

Xpath: //span[text()='BMW Brilliance Real Time Monitoring']

```
<html xmlns="http://www.w3.org/1999/xhtml">
  <head>
  <body>
    <div id="login_main" class="login_main" style="margin-top: 20px;">
      <div id="login_logo" class="login_logo">
      <div class="login_text">
        <span>BMW Brilliance Real Time Monitoring</span>
      </div>
```

3.last(): 当前节点的最后一个元素.

//div[@id='login_main']/div[last()]

```
<document>
  <html xmlns="http://www.w3.org/1999/xhtml">
    <head>
    <body>
      <div id="login_main" class="login_main" style="margin-top: 20px;">
        <div id="login_logo" class="login_logo">
        <div class="login_text">
        <div class="login-center">
        <div class="login_bottom">
```

4.starts-with(): //div[starts-with(@id,'login-toggle')], 表示选择以'login-toggle'开头的id属性的div节点

```
<html xmlns="http://www.w3.org/1999/xhtml">
  <head>
  <body>
    <div id="login_main" class="login_main" style="margin-top: 20px;">
      <div id="login_logo" class="login_logo">
      <div class="login_text">
      <div class="login-center">
      <div class="login_bottom">
        <div id="login-toggle-language" class="login-toggle-language">
```

About Mozilla Firefox



Click F12

PRACTICE TO WRITE THE AUTOMATION TEST

Story: Navigate to Baidu, searching 'Cucumber'
As a user want to search Cucumber function

URL: <https://www.baidu.com/>

Searching 'Cucumber'

Click 'Search' button

Feature: Navigate to Baidu, searching 'Cucumber'
As a user want to search Cucumber function

Background:

Given I navigate to "https://www.baidu.com/"

Scenario: Searching 'Cucumber'

When I enter "Cucumber" into input field having id "kw"

And I click on element having id "su"

Create Issue

Project* A_auto_demo1 (AAUT)

Issue Type* Story

Summary*

Component/s

Priority

Reporter*

Assignee

- Test
- Test Set
- Test Execution
- Pre-Condition
- Test Plan

➤ What is Xray?

Xray is a complete Test Management tool for Jira. It supports the entire testing life cycle: test **planning**, test **design**, test **execution** and test **reporting**.

A test case. It may either be manual or automated, composed of multiple steps, actions and expected results.

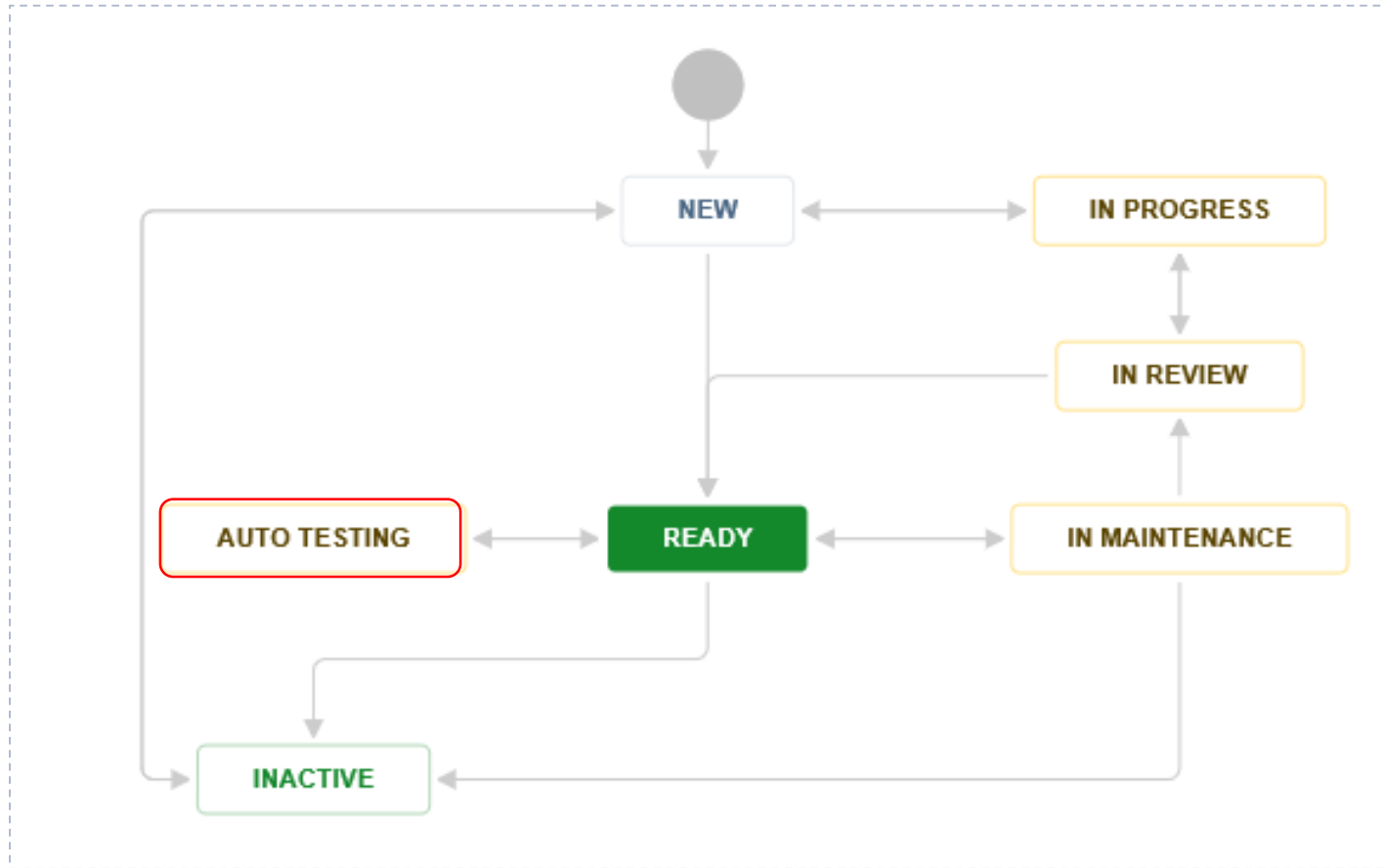
A group of tests, organized in some logical way. A test may belong to multiple Test Sets.

An assignable, "schedulable" task to execute one or more tests for a given version/revision along with its results.

Initial conditions that must be assured before the execution of the Test's own steps. A Pre-Condition may be used by multiple Tests.

A "formal" plan of the tests intended to be executed for a given version.

XRAY TEST WORKFLOW

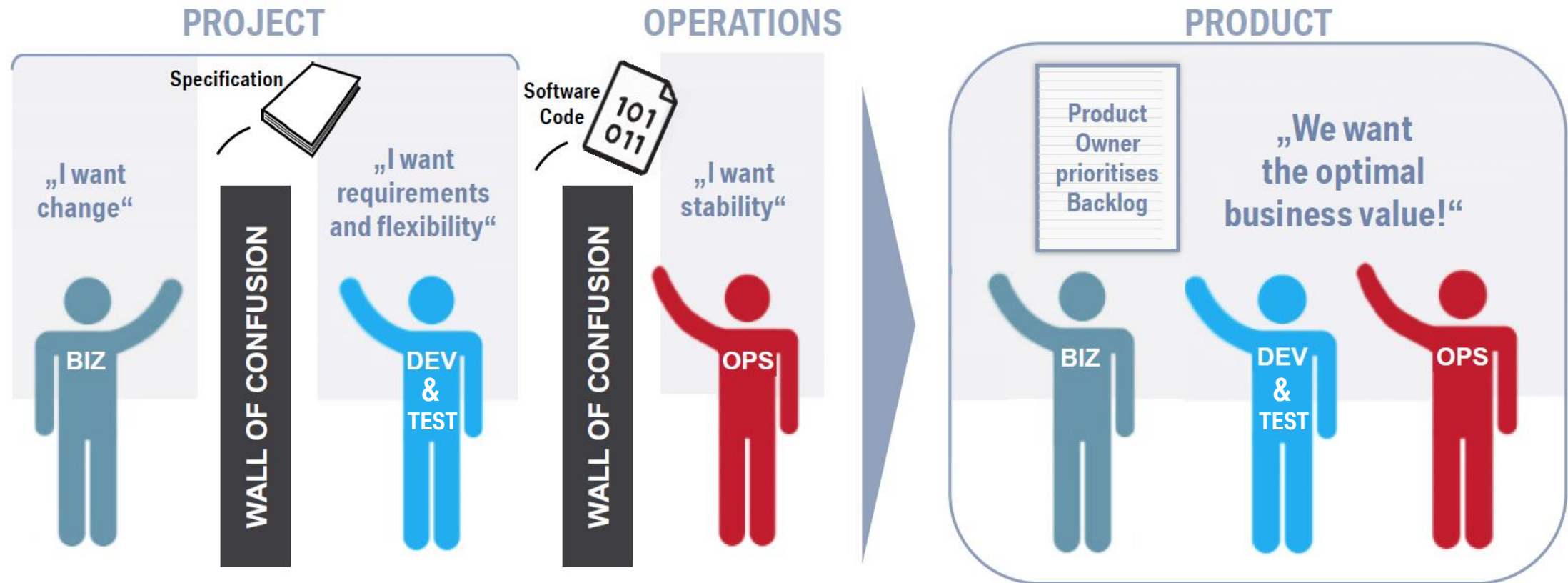


Auto Testing: It's a transient status, after your test run finished, it will change to ready automatically.

Notice: If the test failed, the status change to ready as well, but if the tester want to modify the test, they need to go back to maintenance status.

PROJECT ORGANIZATION.

TARGET: PRODUCTS WITH SKILLS FOR BIZ, DEV, TEST AND OPS.



Product teams with **End2End responsibility** and **BizDevOps** skills.

Automation Output

140.00%

120.00%

100.00%

80.00%

60.00%

40.00%

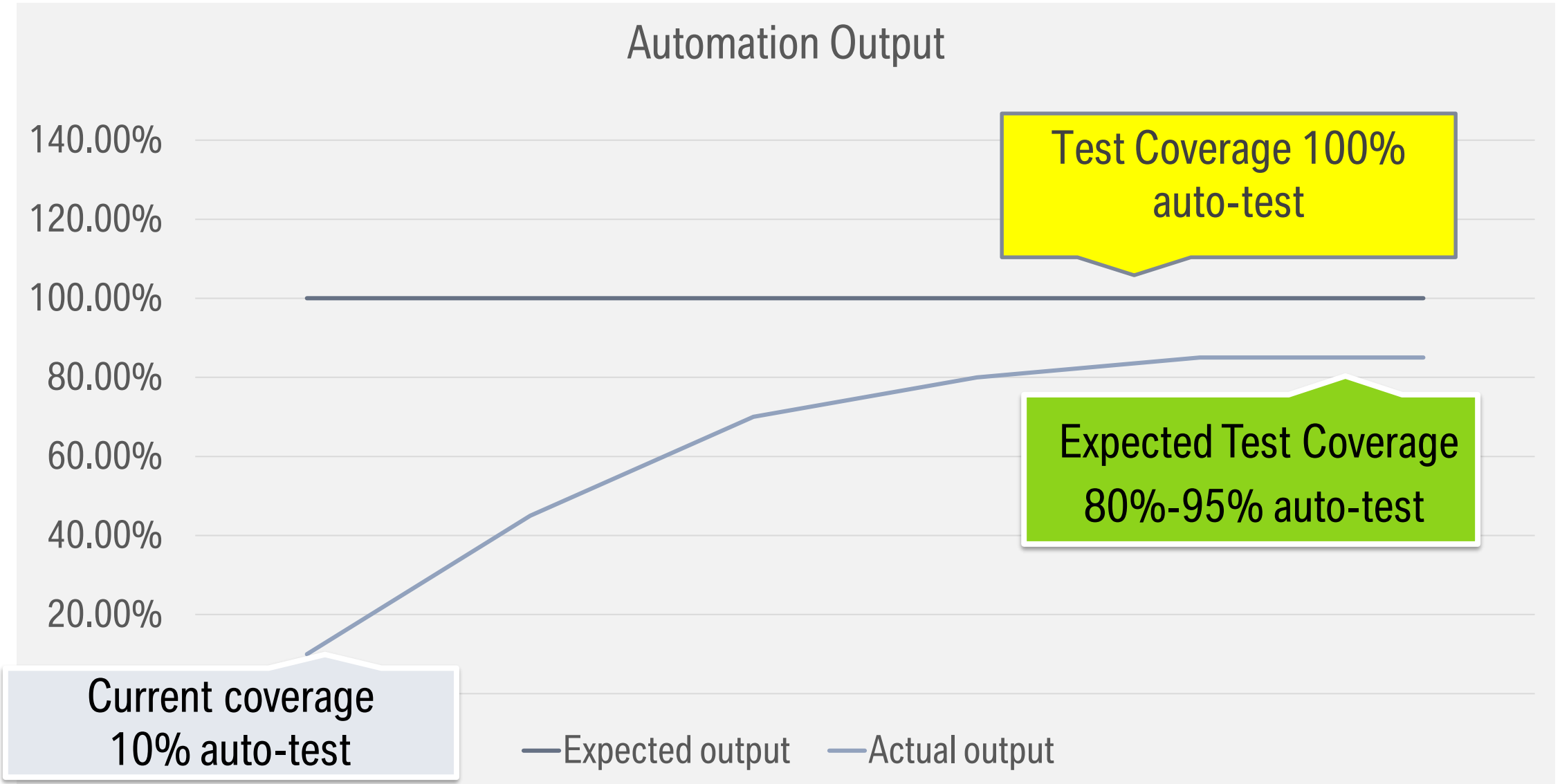
20.00%

Test Coverage 100%
auto-test

Expected Test Coverage
80%-95% auto-test

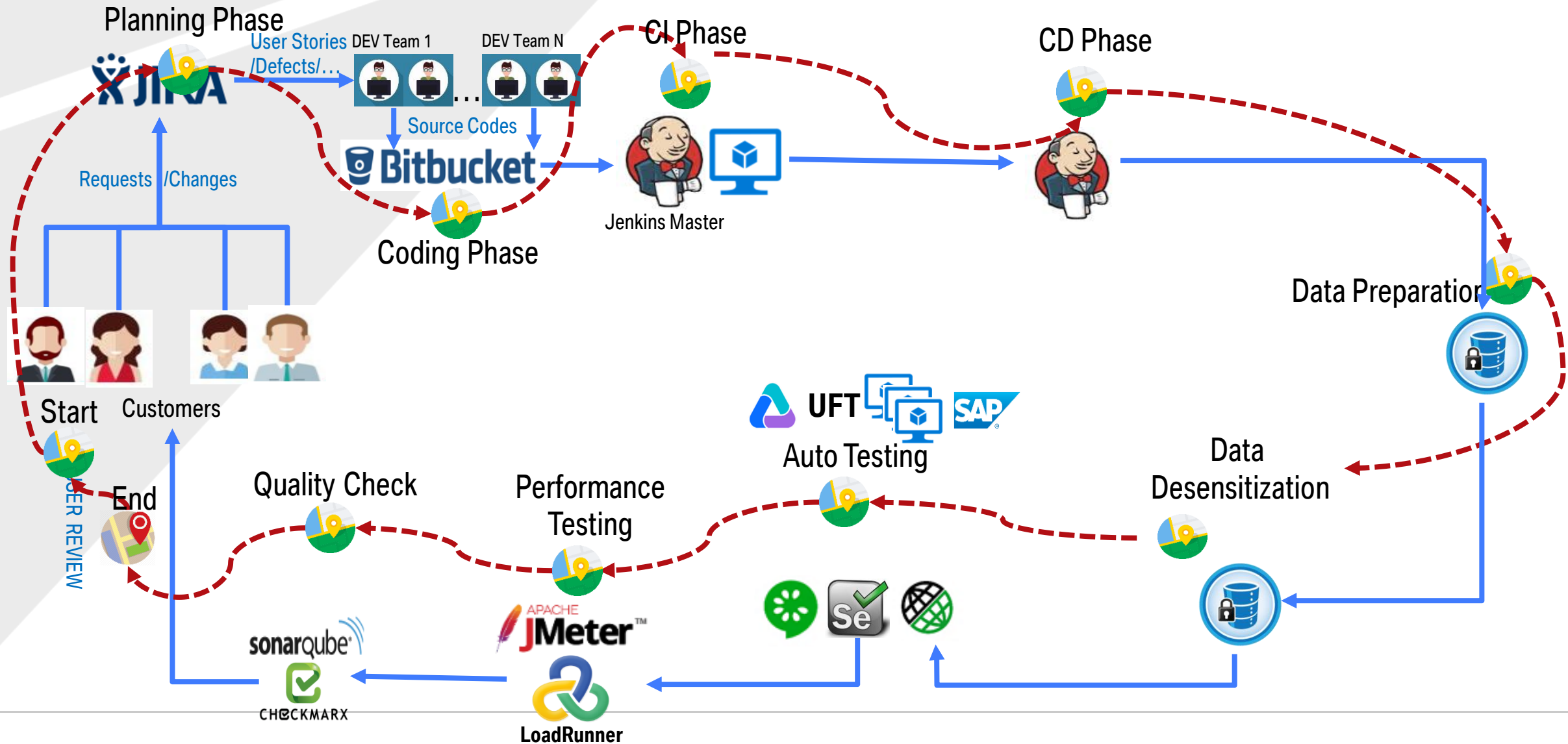
Current coverage
10% auto-test

— Expected output — Actual output



DEVOPS PLATFORM.

HOW TO DO: AUTOMATION TEST INTEGRATE WITH DEVOPS.



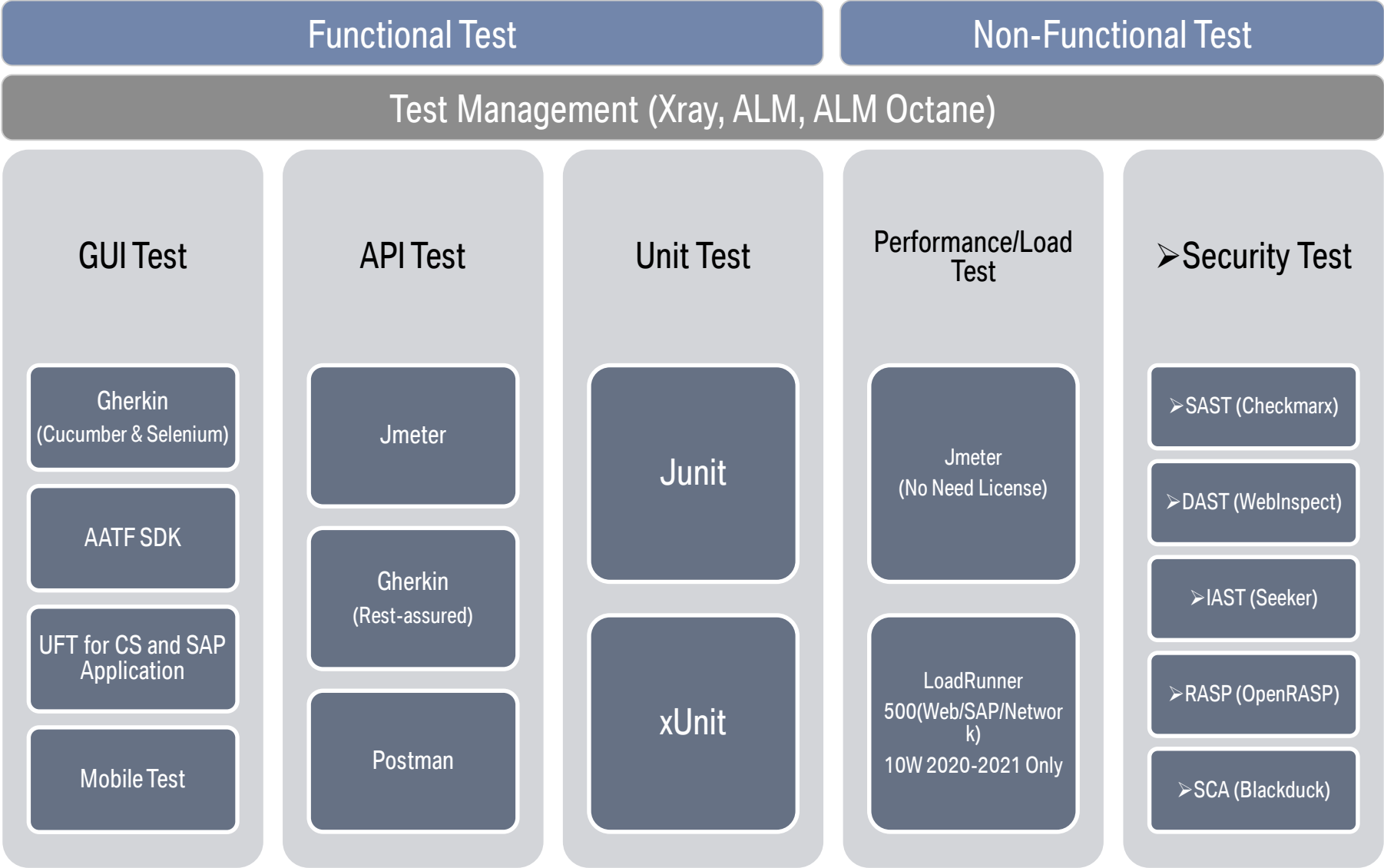
THANKS

Q&A

back up

WHAT CAN WE DO?

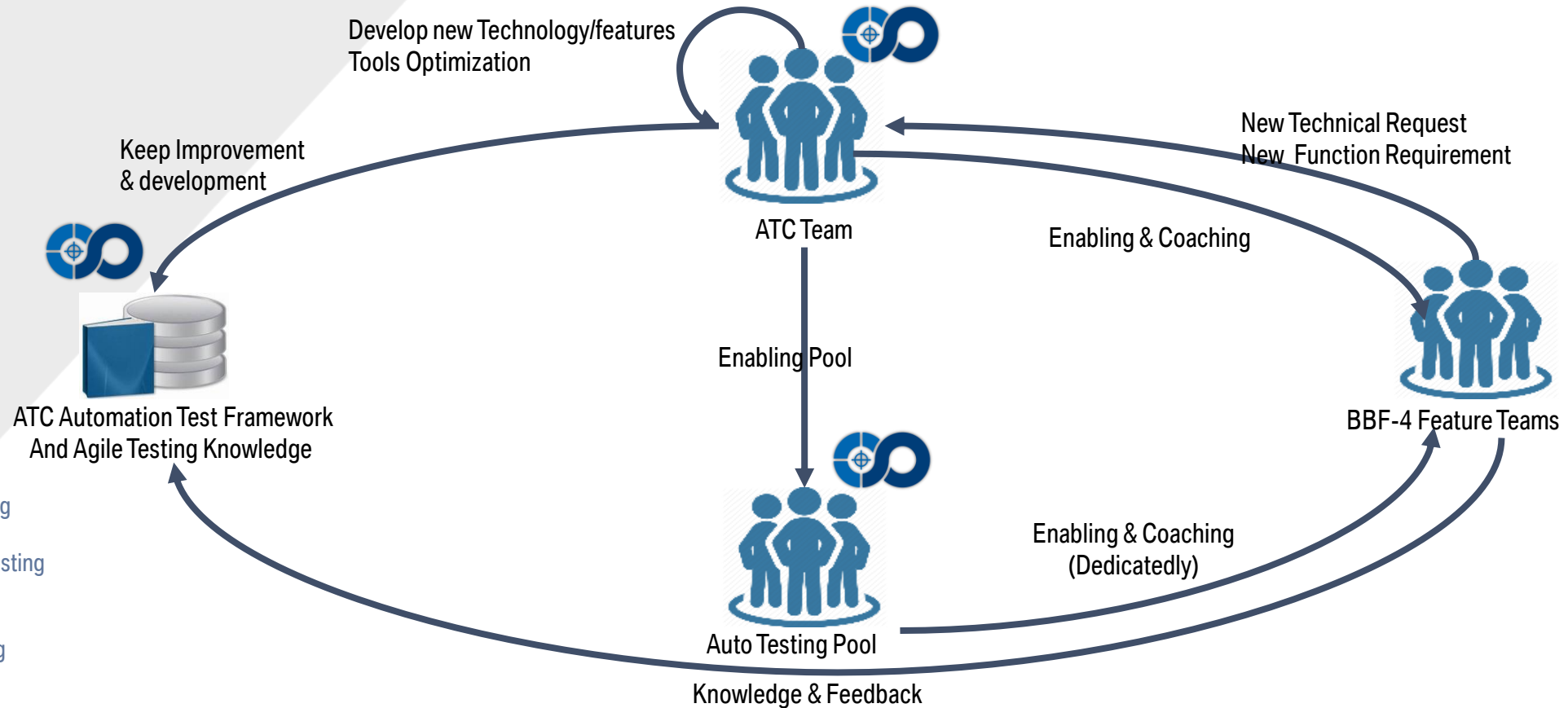
ATC AUTOMATION TEST FRAMEWORK





Vision

“Make everyone become Auto-Test Expert”



- ☐ Function Testing
- ☐ API Testing
- ☐ No-Function Testing
- ☐ Mobile Testing
- ☐ SAP Testing
- ☐ Security Testing
- ☐ Load Testing
- ☐ Etc.

AUTOMATION TEST TOOLS IN BBA ATC

1. Test Management Tools



2. Unit Test

3. Functional Test Tools in ATC(ATC Automation Test Framework)

➤ B/S Application Automation test(Open Source)

❑ GUI



❑ API



➤ C/S OR C/S + B/S+ Mobile (ATC Provide License)

❑ UFT



4. Non-Functional Test

➤ Performance/Load Test



➤ Security Test



Seeker
Interactive Application Security Testing

Checkmarx

MICRO FOCUS
Webinspect

Open source RASP

BLACKDUCK